

Enro 10% Liquid Solution

Liquid Solution
(FOR VETERINARY USE ONLY)

COMPOSITION

Enrofloxacin.....100g/L

PRODUCT DESCRIPTION

The finished product is colorless to yellowish transparent fluid liquid

PHARMACODYNAMIC

Enrofloxacin is a bactericidal agent. The bactericidal activity of enrofloxacin is concentration dependent, with susceptible bacteria cell death occurring within 20-30 minutes of exposure. Enrofloxacin has demonstrated a significant post-antibiotic effect for both Gram positive and Gram negative bacteria and is active in both stationary and growth phases of bacterial replication. Its mechanism of action is not thoroughly understood, but it is believed to act by inhibiting bacterial DNA-gyrase (a type-II topoisomerase), thereby preventing DNA super coiling and DNA synthesis. Both enrofloxacin and ciprofloxacin have similar spectrums of activity. These agents have good activity against many gram negative bacilli and cocci, including most species and strains of *Pseudomonas aeruginosa*, *Klebsiella* spp., *E. coli*, *Enterobacter*, *Campylobacter*, *Shigella*, *Salmonella*, *Aeromonas*, *Haemophilus*, *Proteus*, *Yersinia*, *Serratia*, and *Vibrio* species. Of the currently commercially available quinolones, ciprofloxacin and enrofloxacin have the lowest MIC values for the majority of these pathogens treated. Other organisms that are generally susceptible include *Brucella* sp, *Chlamydia trachomatis*, *Staphylococci* (including penicillin producing and methicillin-resistant strains), *Mycoplasma*, and *Mycobacterium* sp. (not the etiologic agent for Johne's Disease). The fluoroquinolones have variable activity against most *Streptococci* and are not usually recommended to be used for these infections. These drugs have weak activity against most anaerobes and are ineffective in treating anaerobic infections. Resistance does occur by mutation, particularly with *Pseudomonas aeruginosa*, *Klebsiella pneumonia*, *Acinetobacter* and enterococci, but plasmid-mediated resistance is not thought to occur.

PHARMACOKINETIC

The pharmacokinetics of enrofloxacin are such that oral and parenteral administration leads to similar serum levels. Enrofloxacin possesses a high distribution volume. Tissue levels 2-3 higher than that found in the serum, have been demonstrated in laboratory animals and target species. Organs in which high levels can be expected are the lungs, liver, kidney, bone and lymphatic system. Enrofloxacin also distributes into the cerebrospinal fluid, the aqueous humour and the foetus in pregnant animals.



SHENNONG ANIMAL HEALTH (M) SDN. BHD. (441647-X)

No. 2, 4, 6, 8 & 10, Jalan Industri USJ 1/19, Taman Perindustrian USJ 1, 47610 Subang Jaya, Selangor, Malaysia.

INDICATIONS

For use in the treatment of diseases of the respiratory and alimentary tracts of bacterial or mycoplasmal origin (e.g. pasteurellosis, mycoplasmosis, coli-bacillosis, coli-septicaemia and salmonellosis) where clinical experience, supported where possible by sensitivity testing of the causal organism, indicates enrofloxacin as the drug of choice.

CONTRA-INDICATIONS

The product should not be used for prophylaxis

ADMINISTRATION & DOSAGE

The product is administered via the drinking water. Medicated water should be made up immediately prior to provision. Medication of the water supply should be continuous during the treatment period and no other source of water should be available. The product should be given over a minimum period of 6 hours per day to allow all birds to drink.

Poultry and Turkeys : The dose rate is 10 mg/kg bodyweight daily for 3 to 10 days.

Each ml contains 100 mg enrofloxacin. Calculate the daily quantity (ml) of product required for treatment period as follows :

Total number of birds x Average body weight in kg x 0.1 = Total volume (ml) per day

The product may be put directly into the header tank or introduced via a water proportioner pump. For the treatment of salmonellosis or pasteurellosis a treatment period of 5 to 10 days must be used. For other conditions the duration of treatment may be 3 to 5 days.

INTERACTION WITH OTHER MEDICAMENTS

Antacids containing cations (Mg⁺⁺, Al⁺⁺⁺, Ca⁺⁺) may bind to enrofloxacin/ ciprofloxacin and prevent its absorption. Sucralfate may inhibit absorption of enrofloxacin/ ciprofloxacin, separate doses of these drugs by at least 2 hours. Enrofloxacin/ ciprofloxacin administered with theophylline may increase theophylline blood level. Probenecid blocks tubular secretion of enrofloxacin/ ciprofloxacin and may increase its blood level and half-life. Synergism may occur, but is not predictable, against some bacteria (particularly *Pseudomonas aeruginosa* or other Enterobacteriaceae) with these compounds and aminoglycosides, 3rd generation cephalosporins agents, and extended-spectrum penicillins. Although enrofloxacin/ ciprofloxacin has minimal activity against anaerobes, in vitro synergy has been reported when used with clindamycin against strains of *Peptostreptococcus*, *Lactobacillus* and *Bacteroids fragilis*. Nitrofurantoin may antagonize the antimicrobial activity of the fluoroquinolones and their concomitant use is not recommended. Fluoroquinolones may exacerbate the nephrotoxicity of cyclosporine (used systemically). Because the fluoroquinolones are relatively new additions to the therapeutic armamentarium, more interactions may be forthcoming.



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STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Not applicable, the product is indicated only for use in poultry. Do not administer to layer replacement birds within 14 days of coming into lay.

SIDE EFFECT

With the exception of potential cartilage abnormalities in young animals (see Contraindications above), the adverse effect profile of enrofloxacin is usually limited to GI distress (vomiting, anorexia).

OVERDOSE AND TREATMENT

Do not exceed the recommended dose.

WITHDRAWAL TIME : For slaughter withdrawal, 28 days. Eggs: Contraindicated (not for use in birds producing eggs for human consumption)

Warning:

Use under Veterinarian recommendation. Official and local antimicrobial policies should be taken into account when the product is used. Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials. Whenever possible, fluoroquinolones should only be used based on susceptibility testing. Use of the product deviating from the instructions given in the SPC may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Keep out of reach of children

SHELF LIFE : 24 months from manufacturing date. After first opening of container, please keep it tight and use within 3 months. And finish treatment within 12 hours after dilution/ mixing.

STORAGE : Store in a cool & dry place ($\leq 30^{\circ}\text{C}$). Strictly avoid direct sunlight & heat exposure.

PACKAGING :Product is filled in plastic bottle of 1000 ml, sealed with aluminum foil and aluminum screw cap, fitted guarantee plastic cap.

DATE OF REVISION: 14 Apr 2019



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